

Daal is now being sold at Rs. 20 a kg. During last month its rate was Rs. 16 per kg. By how much percent should a family reduce its consumption so as to keep the expenditure fixed?

☒ 22%

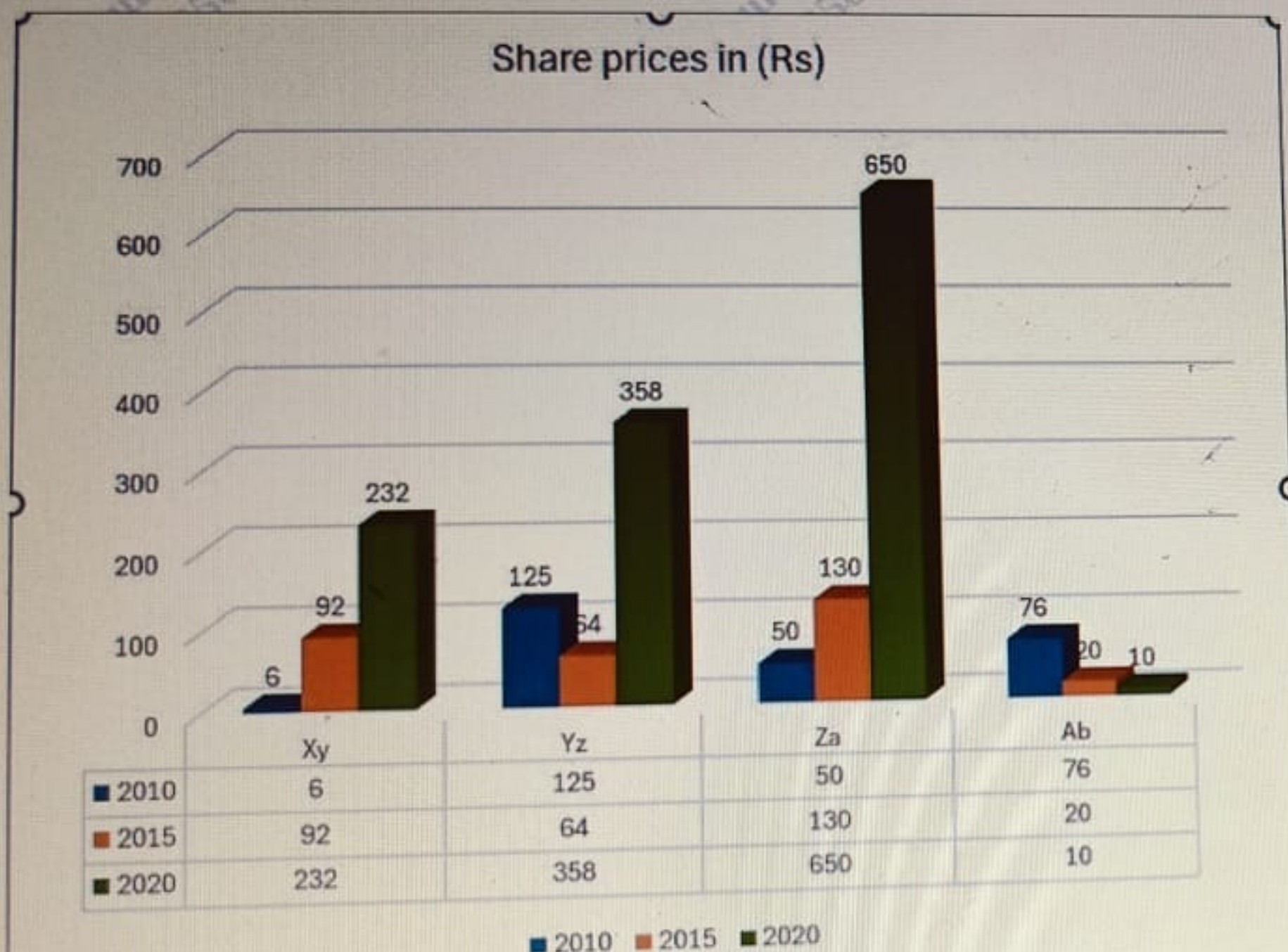
☐ 32%

☐ 30%

☐ 20% ✓

3 Direction: Read the information carefully and answer the questions given below

4 The below bar chart gives information about the share prices (in Rs) of four shares in 3 different years.



15 Question: The share price of "Za" is estimated to increase by 200% from 2020 to 2022. What will be the share price of "Za" in the year 2022 as per the estimation?

16 ☐ Rs. 1950 ✓

17 ☐ Rs. 2700

18 ☐ Rs. 2400

19 Previous Question

Next Question



QUESTION: 3 GROUP: Aptitude SECTION: Data Analysis Mark(s): 1

Direction: Read the information carefully and answer the questions given below

The below bar chart gives information about the share prices (in Rs) of four shares in 3 different years.



Question: Which share has seen decline in its price from 2010 to 2020?

☒ Ab

☐ Xy

☐ Yz

Previous Question

Next Question

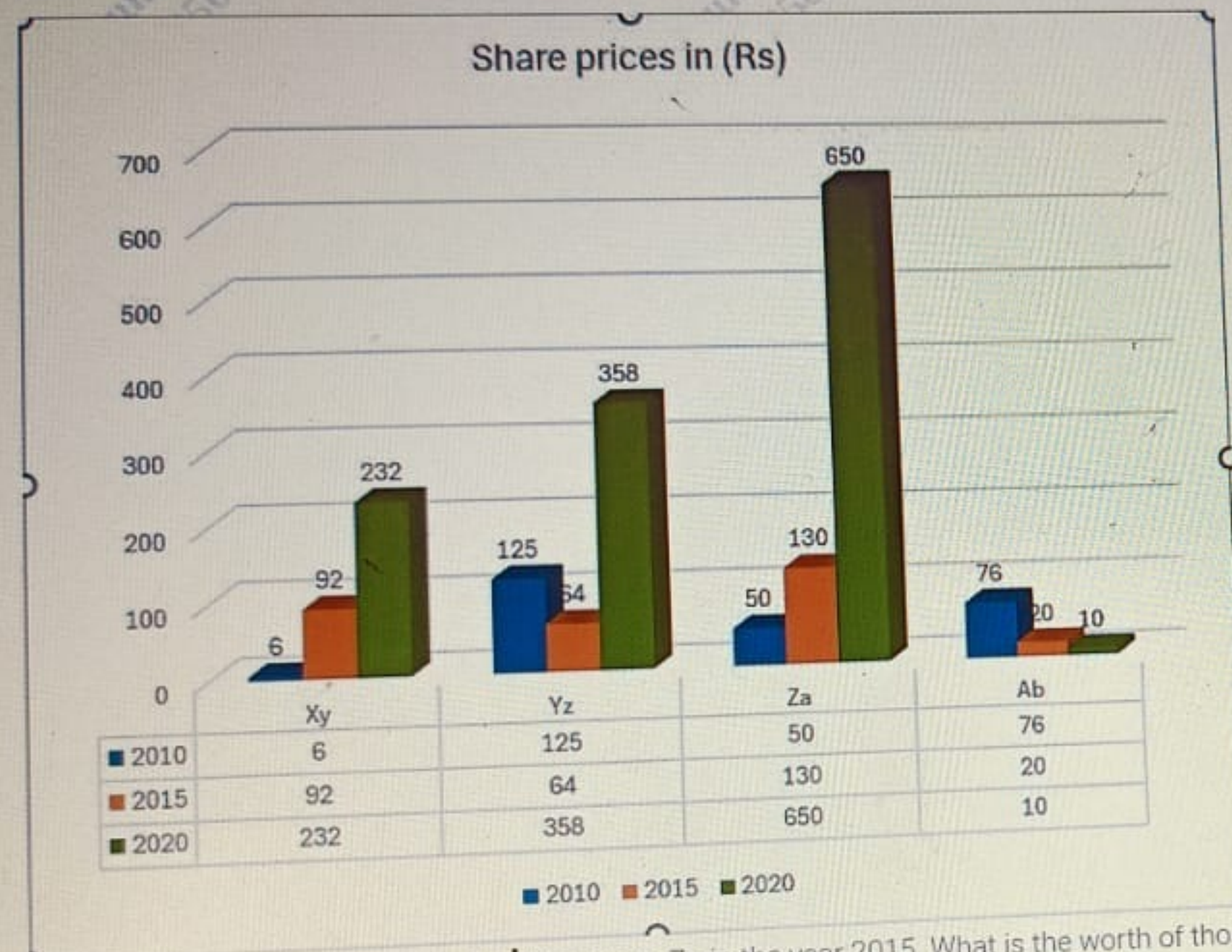


1
2
3
4 ☆
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QUESTION: 4 GROUP: Aptitude SECTION: Data Analysis Mark(s): 1

Direction: Read the information carefully and answer the questions given below

The below bar chart gives information about the share prices (in Rs) of four shares in 3 different years.



Question: Alex has invested Rs. 39000 in share Za in the year 2015. What is the worth of those shares in the year 2020?

- ☐ Rs. 78,000
- ☐ Rs. 36,000
- ☐ Rs. 1,95,000

Previous Question

Next Question



QUESTION: 5 GROUP: Aptitude SECTION: Data Analysis Mark(s): 1

Direction: Read the information carefully and answer the questions given below

The below bar chart gives information about the share prices (in Rs) of four shares in 3 different years.



Question: For which of the following shares, the percentage increase in the share price is maximum from 2015 to 2020?

☐ Ab

☐ Xy

☒ Yz

Previous Question

Next Question

A box contains 10 good articles and 6 with defects. One article is chosen at random. What is the probability that it is either good or has a defect?

☐ 5/8

☐ 3/8

☒ 1

☐ 3/4

In 10 seconds, a 125 m long train passes a man running at 5 km/hr in the same direction of the train. The speed of the train is:

☐ 54 km/hr

☐ 55 km/hr

☒ 50 km/hr ✓

☐ 45 km/hr

The integers a_n and b_n are defined by the relationship $(a_n + b_n\sqrt{2}) = (1 + \sqrt{2})^n$ for all integers $n \geq 1$. What is the value of $\text{HCF}(a_n, b_n)$ for all integers $n \geq 1$?

- ☐ 1
- ☐ Undefined
- ☐ $\sqrt{2}$
- ☐ $2^{(3n)}$

The marked price of a selfie stick is 100% above the cost price. When marked price is increased by 25% and selling price is increased by 25%, the profit doubles. If original marked price is Rs. 600, then what is the

☐ Rs. 450

☐ Rs. 350

☒ Rs. 400 ✓

☐ Rs. 550

Attitude SECTION: Problem Solving Mark(s): 1

A stick is 100% above the cost price. When marked price is increased by 25% and selling price is increased by 25%, the profit doubles. If original marked price is Rs. 600, then what is the original selling price?

Attitude SECTION: Problem Solving Mark(s): 1

A stick is 100% above the cost price. When marked price is increased by 25% and selling price is increased by 25%, the profit doubles. If original marked price is Rs. 600, then what is the original selling price?

QUESTION: 10 GROUP: Aptitude SECTION: Problem Solving Mark(s): 1

For a set of 6 MCQ questions in an interview which had four choices, none of the candidate had written all correct answers and no candidates gave the same sequence of answers. Find the maximum number of candidates who attempted the 6 questions.

☐ 1296 ways

☒ 4095 ways ✓

☐ 720 ways

☐ 360 ways

one of the candidate had written all correct answers and no candidates gave the same sequence of answers. Find the maximum number of candidates who attended the interview, if everyone attempted all

Direction: Study the information given below and answer the questions that follow.

Five friends Alice, Ben, Clara, David, and Eva participate in a 10 km charity run. Each friend runs at a different constant speed. The following information is known:

- Alice completes the race in 50 minutes.
- Ben runs 1 kilometer in 6 minutes.
- Clara runs at a speed of 8 kilometers per hour.
- David takes 1 hour and 10 minutes to finish the race.
- Eva runs at a speed 2 kilometers per hour faster than Ben.

Question: Who among the following finishes the race first?

☐ Clara

☐ Ben

☐ David

☒ Alice



Direction: Study the information given below and answer the questions that follow.

Five friends Alice, Ben, Clara, David, and Eva participate in a 10 km charity run. Each friend runs at a different constant speed. The following information is known:

- Alice completes the race in 50 minutes.
- Ben runs 1 kilometer in 6 minutes.
- Clara runs at a speed of 8 kilometers per hour.
- David takes 1 hour and 10 minutes to finish the race.
- Eva runs at a speed 2 kilometers per hour faster than Ben.

Question: Who among the following has the highest running speed?

☒ Eva

☐ Ben

☐ David

☐ Clara

Direction: Study the information given below and answer the questions that follow.

Five friends Alice, Ben, Clara, David, and Eva participate in a 10 km charity run. Each friend runs at a different constant speed. The following information is known:

- Alice completes the race in 50 minutes.
- Ben runs 1 kilometer in 6 minutes.
- Clara runs at a speed of 8 kilometers per hour.
- David takes 1 hour and 10 minutes to finish the race.
- Eva runs at a speed 2 kilometers per hour faster than Ben.

Question: How long does it take Clara to complete the race?

☒ 1.25 hours ✓

☐ 1.16 hours

☐ 0.83 hours

☐ 1 hour

Direction: Study the information given below and answer the questions that follow.

Five friends Alice, Ben, Clara, David, and Eva participate in a 10 km charity run. Each friend runs at a different constant speed. The following information is known:

- Alice completes the race in 50 minutes.
- Ben runs 1 kilometer in 6 minutes.
- Clara runs at a speed of 8 kilometers per hour.
- David takes 1 hour and 10 minutes to finish the race.
- Eva runs at a speed 2 kilometers per hour faster than Ben.

Question: Who would finish the race last if Eva slowed down by 2 kilometers per hour?

☒ Clara

☐ Ben

☐ Alice

☐ David

Direction: Study the information given below and answer the questions that follow.

Five friends Alice, Ben, Clara, David, and Eva participate in a 10 km charity run. Each friend runs at a different constant speed. The following information is known

- Alice completes the race in 50 minutes.
- Ben runs 1 kilometer in 6 minutes.
- Clara runs at a speed of 8 kilometers per hour.
- David takes 1 hour and 10 minutes to finish the race.
- Eva runs at a speed 2 kilometers per hour faster than Ben.

Question: If David had run 1 kilometer per hour faster, how much time would he have saved?

☒ 0.12 hours

☐ 0.8 hours

☐ 1.2 hours

☐ 0.6 hours

Pointing to a photograph Preethi said "She is my father's sister's daughter Vanitha", how is Preethi's father related to Vanitha?

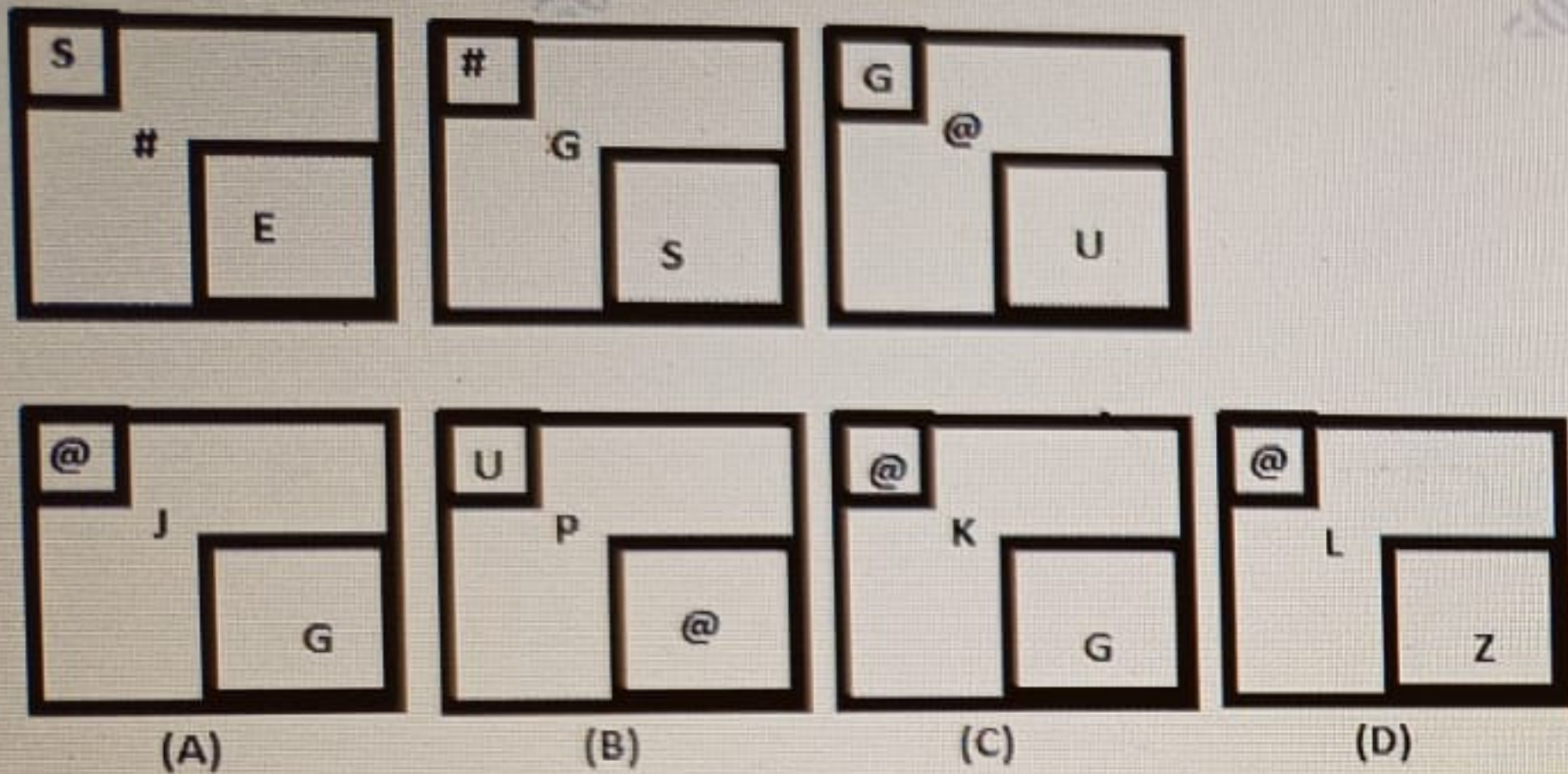
☐ Father

☐ Grandfather

☒ Uncle

☐ Cousin

A set of problem figures and answer figures are given below, in problem figure the images are related in some way. Choose the correct figure from answer figure to complete the series.



☐ (A)

☐ (B)

☐ (C)

☐ (D)

QUESTION: 18 GROUP: **Aptitude** SECTION: **Problem Solving** Mark(s): 1

I have a 60 cm long gold necklace with one joint. If I break the necklace in two pieces (by giving two independent random cuts), which of the following is closest to the expected length of the piece

☐ 20 cm

☐ 25 cm

☐ 40 cm

☐ 30 cm

. If I break the necklace in two pieces (by giving two independent random cuts), which of the following is closest to the expected length of the piece containing the joint?

QUESTION: 19 GROUP: **Aptitude** SECTION: **Problem Solving** Mark(s): 1

Mary and Maurice can do a piece of work in 10 days and 15 days respectively. They work together for 3 days and then Maurice leaves. Mary finishes the remaining work alone. In how many days

☐ 9 days

☐ 6 days

☐ 10 days

☒ 8 days

Solving

Mark(s): 1

5 days respectively. They work together for 3 days and then Maurice leaves. Mary finishes the remaining work alone. In how many days is the total work finished?

QUESTION: 20 GROUP: **Aptitude** SECTION: **Puzzles** Mark(s): 1

How many pairs of letters are there in the word '**IMPURE**' which has many letters between them (from both the sides) as in the alphabets from A to Z?

☐ 3

☐ 0

☒ 1



☐ 2

The following code is attempting to calculate and print the count of lonely 1s but it may contain some error. Which of the statements can fix the error (if exists)?

```
#include <stdio.h>
#include <stdlib.h>
#include <string.h>

void CountLonelyOne(int number) {
    char num_str[20];
    sprintf(num_str, "%d", number);
    int count = 0;
    int len = strlen(num_str);

    for (int i = 0; i < len; i++) {
        if (num_str[i] == '1') {
            if (i == 0) {
                if (len == 1 || num_str[i + 1] != '1') {
                    count++;
                }
            } else if (i == len - 1) {
                if (num_str[i - 1] != '1') {
                    count++;
                }
            } else {
                if (num_str[i - 1] != '1' && num_str[i + 1] != '1') {
                    count++;
                }
            }
        }
    }
}
```

```

21 ☆ if (num_str[i] == '1') {
      if (i == 0) {
22         if (len == 1 || num_str[i + 1] != '1') {
23             count++;
24         }
25     } else if (i == len - 1) {
26         if (num_str[i - 1] != '1') {
27             count++;
28         }
29     }
30 }
31 print("%d\n", count);
32 }

```

```

32 int main() {
33     int N;
34     scanf("%d", &N);
35     CountLonelyOne(N);
36     return 0;
37 }

```

- i) The `sprintf(num_str, "%d", number);` line should use `%s` instead of `%d`.
- ii) The `for (int i = 0; i < len; i++)` loop should start from `i = 1` instead of `i = 0`.
- iii) The `print("%d\n", count);` statement should be `printf("%d\n", count);`.
- iv) The `scanf("%d", &N);` line should use `%s` instead of `%d`.

```

        count++;
    }
}
}
print("%d\n", count);
}

```

```

int main() {
    int N;
    scanf("%d", &N);
    CountLonelyOne(N);
    return 0;
}

```

- i) The `sprintf(num_str, "%d", number);` line should use `%s` instead of `%d`.
- ii) The `for (int i = 0; i < len; i++)` loop should start from `i = 1` instead of `i = 0`.
- iii) The `print("%d\n", count);` statement should be `printf("%d\n", count);`.
- iv) The `scanf("%d", &N);` line should use `%s` instead of `%d`.

- ☐ Statement iv
- ☐ Statement i and ii
- ☐ Statement iii
- ☐ Statement i and iii

Previous Question

Next Question

What will be the output of the program given below?

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    int c[ ] = {2.8, 3.4, 4, 6.7, 5};
```

```
    int j, *p = c, *q = c;
```

```
    for (j = 1; j < 5; j++)
```

```
    {
```

```
        printf("%d ", *c);
```

```
        ++q;
```

```
    }
```

```
    for (j = 1; j < 5; j++)
```

```
    {
```

```
        printf("%d ", *p);
```

```
        ++p;
```

```
    }
```

```
    return 0;
```

```
}
```

☐ 2 3 4 6 5

☒ 2 2 2 2 2 3 4 6

☐ 2 2 2 2 2 3 4 6 5

☐ 2.8 3.4 4 6.7 5

21

QUESTION: 23 GROUP: Programming Basics SECTION: Data Structures Basics Mark(s): 1

22

Consider an empty stack S and queue Q as given below:

23 ☆

Q: f → y → u → i

24

What will be the content of stack after the following operations on S and Q?

25

PUSH(x) → to insert an element x in S

26

POP → to delete the top most element of S.

27

ADD(X) → to add an element in Q

28

DEL → to delete an element from Q.

29

I) PUSH(w), ADD(q)

II) DEL, DEL, ADD(e)

30

III) PUSH(DEL), PUSH(o), PUSH(u), PUSH(p)

IV) ADD(b), ADD(m), DEL, POP

31

32

☐ e → o → u → p

33

☐ w → e → o → u

34

☐ u → o → u → p

35

36

☒ w → u → o → u

37

38

[Previous Question](#)[Next Question](#)

21

QUESTION: 24 GROUP: Programming Basics SECTION: Data Structures Basics Mark(s): 1

22

23

Which of the below given statement(s) is/are correct about quick sort?

24 ☆

- I) Quick sort has $O(n \log n)$ time complexity in worst case.
II) Quick sort uses greedy algorithm.

25

26

☐ Neither (I) nor (II) ✓

27

☐ Both (I) and (II)

28

☐ Only (I)

29

☐ Only (II)

30

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38

The given code aims to print the 5th number in an array named numbers, initialized with the values {1, 2, 3, 4, 5}. However, there is an error in the code. Identify the line causing the error.

```
#include <stdio.h> // Line 1

int main() {
    int numbers[5] = {1, 2, 3, 4, 5}; // Line 3
    printf("The 5th number in the array is: %d\n", numbers[5]); // Line 4
    return 0; // Line 5
}
```

☒ Line 4

☐ Line 3

☐ Line 1

☐ Line 5

What will be the output of the program given below? /* Assume address of a is 3651*/

```
#include <stdio.h>
int main()
{
    int a;
    int *p;
    int **q;
    a = 30;
    p = &a;
    q = &p;
    printf("a = %d, p = %d and q = %d", a, *p, **q);
    return 0;
}
```

☐ a = 3651, p = 30 and q = 3801

☐ a = 30, p = 3651 and q = 3801

☐ a = 30, p = null and q = null

☒ a = 30, p = 30 and q = 30

22

QUESTION: 27 GROUP: Programming Basics SECTION: Computer Architecture Mark(s): 1

23

24

What is the resultant value after performing a NOT operation on the given binary number?

25

10011011

26

☐ 11111111

27 ☆

☐ 10011011

28

29

☐ 10101100

30

☐ 01100100

31

32

33

34

35

36

37

QUESTION: 20 GROUP: Programming Basics SECTION: Operating System Fundamentals Mark(s): 1

I) Heap allocation

II) Stack allocation

☐ Neither (I) nor (II)

☐ Only (I)

☐ Both (I) and (II)

☒ Only (II)

Convert the following binary number to Decimal

$$(11111111)_2 = ?$$

☒ $(255)_{10}$

☐ $(355)_{10}$

29



Flag this question

 $)_{10}$

☐ $(256)_{10}$

Which one of the given options is the minimum number of AND gates required for the below given?

PZ + NZ

☐ 0

☒ 1

☐ 2

☐ 3

QUESTION: 31

GROUP: Programming Basics

SECTION: C Programming

Mark(s): 1

What will be the output of the program given below?

```
#include <stdio.h>
int main()
{
    unsigned int f = 4, i;
    for (i = 0; i < 2; i++)
    {
        f <= ++i ^ f - 1;
        if (f > 2)
            printf("%d ", f ^ i);
        else
            printf("%d ", f & i);
    }
    return 0;
}
```

☐ 0☒ 17☐ 10☐ 16

What will be the output of the program given below?

```
#include <stdio.h>

int main()
{
    static int p = 2, q = 3;
    p = p << 2 ^ q - 1;
    q = q ^ p;
    printf("%d", p & q);
    return 0;
}
```

☐ 5☐ 6☐ 10☒ 8

Using SJF pre - emptive scheduling algorithm, what will be the average waiting time of the processes given below?

Process	Arrival Time(ms)	Burst Time(ms)
P ₁	0.0	7
P ₂	2.0	4
P ₃	4.0	1
P ₄	5.0	4

☐ 5 ms

☐ 4 ms

☒ 3 ms

☐ 2 ms

What is the binary value of the given hexadecimal number?

$(A1B23)_{16}$

☐ $(0101\ 0110\ 0111\ 1011\ 0011)_2$

☐ $(1101\ 0110\ 0111\ 1011\ 0011)_2$

☒ $(1010\ 0001\ 1011\ 0010\ 0011)_2$ ✓

☐ $(1111\ 0110\ 0111\ 1011\ 0011)_2$

23

24

Which order of operations given in options when performed on an empty stack of size 4 will have TOP $\rightarrow$ L?

25

Operations

26

I) PUSH(A)

27

II) POP()

28

III) PUSH(X)

IV) PUSH (Y)

V) PUSH(L)

29

30

☐ (II), (I), (III), (IV), (V), (II), (III)

31

☐ (I), (III), (II), (II), (V), (II), (II)

32

☐ (V), (IV), (II), (I), (V), (III), (IV)

33

☒ (I), (II), (III), (I), (V), (IV), (II)

34

35

36

QUESTION: 36

GROUP: Programming Basics

SECTION: Data Structures Basics

Mark(s): 1

The leaf node of a tree is also called as

☐

Internal Node

☐

Predecessor Node

☒

Terminal Node

☐

Root Node

What does control flow refer to in C programming?

- ☐ The rate of program execution
- ☒ The order in which statements are executed
- ☐ The structure of a program's functions
- ☐ The flow of data in the program

QUESTION: 38

GROUP: Programming Basics

SECTION: Computer Architecture

Mark(s): 1

What is the hexadecimal equivalent of the given binary number?

11011100110001

☒ 3731h

☐ 9774h

☐ 1800h

☐ 1589h

QUESTION: 39

GROUP: Programming Basics

SECTION: Operating System Fundamentals

Mark(s): 1

Which of the scheduler given in options is also called as the job scheduler?

☐ Short Term Scheduler

☒ Long Term Scheduler

☐ Medium Term Scheduler

☐ Intermediate Term Scheduler

QUESTION: 40 GROUP: **Programming Basics** SECTION: **C Programming** Mark(s): 1

What is the significance of the declaration "const char* p"?

- ☐ The given declaration is an invalid one.
- ☐ The character referenced by p is a constant.
- ☒ p is a constant and not the character pointed by p. ✓
- ☐ Both p and the character pointed by p are constant.

41 ☆

QUESTION: 41 GROUP: Computer Science SECTION: C programming Mark(s): 1

42

What will be the output of the program given below?

43

44

#include <stdio.h>

int fun(char, char);

45

int main(void)

{

46

const unsigned char a = 255;

const signed char b = 255;

47

int c;

c = fun(a, b);

48

printf("\n%d %d %d", a, b, c);

49

return 0;

}

50

int fun(char a, char b)

{

51

printf("%d ", a << 1);

printf("%d ", b << 1);

52

}

53

-2 -2

54

255 -1 0

55

-1 -1

56

255 -1 0

57

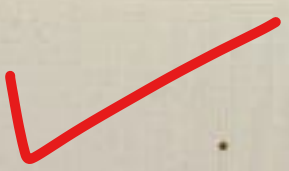
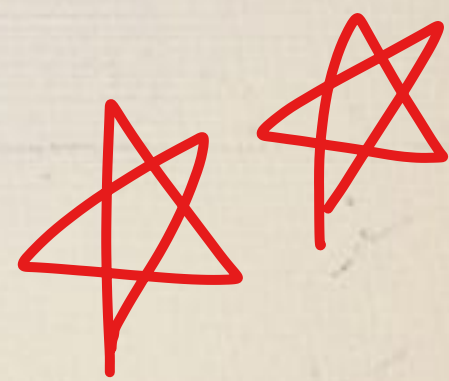
-2 -2

58

255 -1 4

59

Compile time error



41 ☆

What will be the output of the program given below?

42

```
#include <stdio.h>
```

43

```
int fun(char, char);
```

```
int main(void)
```

44

```
{
```

```
    const unsigned char a = 255;
```

45

```
    const signed char b = 255;
```

```
    int c;
```

46

```
    c = fun(a, b);
```

47

```
    printf("\n%d %d %d", a, b, c);
```

```
    return 0;
```

48

```
}
```

```
int fun(char a, char b)
```

49

```
{
```

```
    printf("%d ", a << 1);
```

50

```
    printf("%d ", b << 1);
```

51

```
}
```

52

```
-2 -2
```

53

```
255 -1 0
```

54

```
-1 -1
```

```
255 -1 0
```

55

```
-2 -2
```

56

```
255 -1 4
```

57

```
Compile time error
```

58

59



41

QUESTION: 42 GROUP: Computer Science SECTION: Computer Architecture 1 Mark(s): 1

42 ☆

What is the value of the relative address field of the instruction in decimal if the relative mode branch type is stored in a memory at an address equivalent to the decimal 655 and the branch which is made to an address equivalent to the decimal 500?

43

44

☐ 251

45

☐ 151

46

☐ 250

47

☒ -155 ✓

48

49

50

51

52

53

54

55

56

57

58

59

$$\begin{array}{r} 500 \\ -655 \\ \hline -155 \end{array}$$

GROUP: **Computer Science** SECTION: **Computer Architecture 1** Mark(s): 1

Value of the relative address field of the instruction in decimal if the relative mode branch type is stored in a memory at an address equivalent to the decimal 655 and the branch which is made to an address equivalent to 500?

3

41

QUESTION: 43 GROUP: Computer Science SECTION: Operating Systems Mark(s): 1

42

43 ☆

Which of the below given statement(s) is/are correct about SJF scheduling algorithm?

44

I) This algorithm can cause a problem of starvation.

45

II) This algorithm will always give the minimum average waiting time.

46

☒ Both (I) and (II)

47

☐ Only (II)

48

☐ Only (I)

49

☐ Neither (I) nor (II)

50

51

52

53

54

55

41

QUESTION: 44 GROUP: Computer Science SECTION: C programming Mark(s): 1

42

43

44 ☆



45

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55

Which control flow structure allows the program to repeat a certain block of code as long as a specified condition is true?

☒ for loop ✓

☐ if statement

☐ else statement

☐ switch statement

41

QUESTION: 45

GROUP: Computer Science

SECTION: C programming

Mark(s): 1

42

For the value of abc as 16, what will be printed by the code snippet given below?

43

44

```
unsigned int abc;
```

```
int *to = (int*)&abc;
```

```
printf("%d. ", ++*to++);
```

45 ☆

46

☐ Some garbage value will be printed.

47

☒ 17

48

☐ 18

49

50

☐ Compile time error

51

52

53

54

QUESTION: 46 GROUP: Computer Science SECTION: C programming Mark(s): 1

Which of the line generate(s) error(s) in the program given below?

```
#include <stdio.h>
int main()
{
    int p, *q;
    p = 5;
    q = &p;
    printf("%d ", sink(q, p));
    return 0;
}

int sink(int m, int *n)
{
    int **y;
    y = &n; //Statement 1
    y += m + n; //Statement 2 ✓
    return **y; //Statement 3
}
```

Statement 1 and Statement 2

All Statement 1, Statement 2 and Statement 3

Statement 2 ✓

Statement 3

42

QUESTION: 47 GROUP: Computer Science SECTION: Computer Architecture 1 Mark(s): 1

43

44

Choose the correct option which satisfies the concept of Dynamic Pipeline Scheduling:

45

I) Scoreboarding

46

II) Compiler Scheduling

47 ☆

III) Forwarding

IV) Reorganizing

48

☐ Only (I) and (II)

49

☐ Only (IV)

50

☐ Only (II) and (III)

51

52

☐ Only (I)

53

54

55

56

57

58

59

60

QUESTION: 48 GROUP: Computer Science SECTION: C programming Mark(s): 1

What will be the output of the program given below?

```
#include <stdio.h>

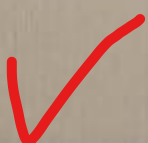
int main()
{
    char p = '1';
    int s = (char *)&p;
    switch(s)
    {
        case '1' : printf("A");
                    break;
        case '0' : printf("B");
                    break;
        default : printf("C");
    }
    return 1;
}
```

1

A

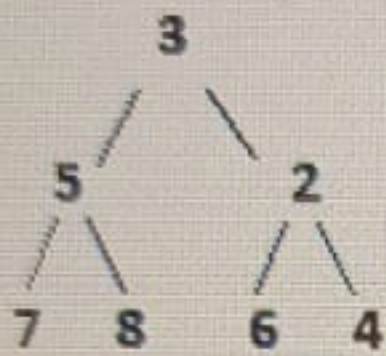
B

C



QUESTION: 49 GROUP: Computer Science SECTION: Data Structures Mark(s): 1

What will be the inorder traversal of the tree given below?



☐ 7 8 5 3 6 4 2

☐ 2 3 4 5 6 7 8

☒ 7 5 8 3 6 2 4

☐ 2 3 5 7 8 6 4

QUESTION: 50 GROUP: Computer Science SECTION: OOPS Mark(s): 1

Which of the statement(s) given below is/are false?

I) The scope of a local inner class is restricted to the block it is defined in. T

✓ II) An inner class method can be accessed from outside the block without using any object. F

III) A local class has access to the members of its enclosing class. T

IV) Local inner classes can extend an abstract class or can also implement an interface. T

☒ Only II ✓

☐ Only I and III

☐ Only II and IV

☐ Only IV

QUESTION: 51

GROUP: Computer Science

SECTION: Algorithms

Mark(s): 1

What will be the final value of n and k after going through the algorithm given below?

Algorithm:

Step 1. Start

Step 2. Initialize a , $n = 2$, $k = 2$ Step 3. Set $k = n$ Step 4. Check if $k < 10$ thenStep 5. Compute $n = k + 1$ Step 6. Compute $k = n * k$ Step 7. Print k Step 8. Else Print n

Step 9. Stop

51 ☆



10 5

52

53

6 3



54

8 1

55

14 1

56

57

58

59

60

QUESTION: 52 GROUP: Computer Science SECTION: OOPS Mark(s): 1

Choose the incorrect statement regarding an "Abstract class".

- ☐ We can declare constructor inside an abstract class. T
- ☐ This class allows an instance or static block inside it. T
- ☒ It is mandatory that abstract class variables must be initialized at the time of its declaration. F
- ☐ Abstract class contains abstract methods as well as concrete methods. T

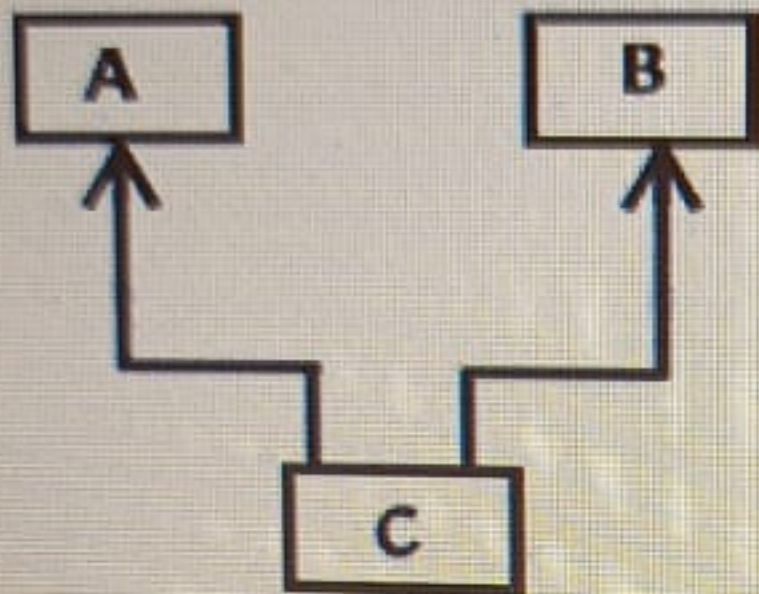
QUESTION: 53

GROUP: Computer Science

SECTION: OOPS

Mark(s): 1

Consider the image given below:



Which of the inheritance types given in the options represents the above diagram?

- ☐ Hybrid Inheritance
- ☒ Multiple Inheritance
- ☐ Multilevel Inheritance
- ☐ Single Inheritance

QUESTION: 54 GROUP: Computer Science SECTION: C programming Mark(s): 1

For swapping values, which one of the given bitwise operator is used efficiently?

☐ |

☒ ^

☐ &

☐ ~

```
x = x ^ y
y = x ^ y
x = x ^ y
```

What is the binary equivalent of the given decimal number?

$(982)_{10}$

☒ 1111010110

☐ 111100110

☐ 110011001

☐ 010111001

QUESTION: 56 GROUP: Computer Science SECTION: C programming Mark(s): 1

What will be the output of the program given below?

```
#include <stdio.h>

int main()
{
    int i, j, k, s;
    int n = 10;
    int a[10] = {1, 24, 24, 65, 65, 98, 56, 89, 89, 0, 0, 56};
    int *p;
    s = n;
    p = a;

    for (i = 0; i < s; i++)
    {
        for (j = 0; j < s; j++)
        {
            if (i == j)
            {
                continue;
            }

            else if (*(p + i) == *(p + j))
            {
                k = j;
                p--;
                while (k == s)
                {
                    *(p + k) = *(p + k - 1);
                    k--;
                }
            }
        }
    }
}
```

42

}

43

else if (*(p + i) == *(p + j))

44

{

k = j;

45

s--;

while (k < s)

46

{

*(p + k) = *(p + k + 1);

47

k++;

}

48

j = 0;

}

49

}

50

}

51

for (i = 0; i < s; i++)

{

52

printf(" %d", a[i]);

}

53

return 0;

54

}

55

1 24 65 98 56 89 0

56 ☆

Run time error

57

Compile time error

58

59

1 24 24 65 65 98 56 89 89 0 0 56

60

What will be the REAR/TAIL and FRONT/HEAD after performing the below given operations on an empty queue of size 3?

Enqueue(7), Enqueue(8), Dequeue(), Enqueue(4), Enqueue(9)

Note: Indexing from 0

☐ REAR = 3 and FRONT = 1

☐ REAR = 1 and FRONT = 2

☒ REAR = 2 and FRONT = 1

☐ REAR = 2 and FRONT = 0

QUESTION: 58 GROUP: Computer Science SECTION: Data Structures Mark(s): 1

Consider a stack having elements : **a b** of capacity 3, what would be the minimum number of **PUSH** and **POP** operations required to overflow the stack?

- ☐ Only 3 POP
- ☐ 3 PUSH and 3 POP
- ☒ Only 2 PUSH
- ☐ 1 PUSH and 3 POP

What will be the physical addresses (or segmentation faults) generated by the given logical addresses if the segmentation system is as below?

	Base	Offset
0	660	199
1	340	106
2	950	50

I) 1, 111 $\rightarrow 111 > 106$ Seg
II) 2, 900 $\rightarrow 900 > 50$

☒ Segmentation fault for both addresses ✓

☐ 217, 950

☐ 451, 1850

☐ 771, Segmentation fault

What is the main reason for the Segmentation fault?

- ☐ Some faults present in the segments.
- ☐ Segments leading to dynamic loading.
- ☒ The part of memory that cannot be accessed.
- ☐ Segments leading to Partitioning.